Faculty of Business and Management
Semester Teaching Plan AY2025-2026 Semester 1

Course Code	GTSC2143
Course Title	Machine Learning for Business Analytics
Course Convener	Lily Bei Luo
Course Teachers	Lily Bei Luo

Week		**Chapter Title / Topic**	**Class Discussion**	**Homework Practice Questions**
1		Course Overview Introduction to Machine Learning Python Basics	Installation of Anaconda Installation of Python Libraries	Distribution of CRA Rubrics to Students
2		Python and Libraries Data Loading Data Preprocessing	Lab Session: Data Munging	Group Formation
3		Data Visualization	Lab Session: Data Munging	Distribution of Individual Assignment Distribution of Team Project
4		Regression I	Lab Session: House Pricing prediction	
5		Regression II	Lab Session: House Pricing prediction	
6		Classification I	Lab Session: Sentiment Analysis	
7		Classification II	Lab Session: Sentiment Analysis	
8		Reading Week		Individual Assignment Submission due
9		Clustering	Lab Session: Document Retrieval	
10		Clustering	Lab Session: Document Retrieval	
11		Recommendation System	Lab Session: Recommentdation System	
12		Team Project Presentation		

Faculty of Business and Management
Semester Teaching Plan AY2025-2026 Semester 1

Course Code	GTSC2143
Course Title	Machine Learning for Business Analytics
Course Convener	Lily Bei Luo
Course Teachers	Lily Bei Luo

13		Team Project Presentation		Oral Presentation Team Project: Submission due by Week 13
14		Catch up and Final Review		

Remarks: Details of Class Discussion and Homework Practice Questions should be included.

Individual Assignment: Questions will be uploaded to iSpace in Week 3 and due in Week 8

Group Assignment: Questions will be uploaded to iSpace in Week 3 and due in Week 13

Detailed Assessment Components (from the approved OBTL course syllabus) and breakdown. (*Assessment tasks and their weightings should not be changed after teaching has begun and the students have been informed)*

Class Participation	10%	
Individual Assignment	20%	
Group Project	30%	
Final Examination (3 hours)	40%	100%

Required Textbook

[1] Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (2nd ed.). O'Reilly Media.
[2] Shalev-Shwartz, S., & Ben-David, S. (2014). Understanding Machine Learning: From Theory to Algorithms. Cambridge University Press.

Reference(s)

[1] Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324
[2] Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. International Conference on Learning Representations (ICLR).
[3] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436–444. https://doi.org/10.1038/nature14539

Course Code	GTSC2143
Course Title	Machine Learning for Business Analytics
Course Convener	Lily Bei Luo
Course Teachers	Lily Bei Luo

[4] Bishop, C. M. (1995). Training with noise is equivalent to Tikhonov regularization. Neural Computation, 7(1), 108–116. https://doi.org/10.1162/neco.1995.7.1.108
[5] Nesterov, Y. (1983). A method of solving a convex programming problem with convergence rate $O(1/k^2)$. Soviet Mathematics Doklady, 27(2), 372–376.
[6] Vapnik, V. N. (1999). An overview of statistical learning theory. IEEE Transactions on Neural Networks, 10(5), 988–999. https://doi.org/10.1109/72.788640
[7] Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 1251–1258.
[8] Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. International Conference on Learning Representations (ICLR).
[9] Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd ed.). Springer.
[10] Schmidhuber, J. (2015). Deep learning in neural networks: An overview. Neural Networks, 61, 85–117. https://doi.org/10.1016/j.neunet.2014.09.003

Prepared by: Dr. Lily Bei LUO

Prepared on: 20 June 2025

Note: Course teachers should release the reviewed STP and rubrics, and the approved course syllabus to students in the first/second week of teaching. Please keep the internal use version of the STP for reference throughout the semester.